Solartron 7075 opto-isolated USB interface

USB-B -> RP2354A -> 6 opto channels -> shift registers -> 50-way D (SKB of the 70754 BCD interface unit)

MCU_USB

File: mcu_usb.kicad_sch

ISOLATION

File: isolation.kicad_sch

DVM_INPUTS

File: dvm_inputs.kicad_sch

DVM_OUTPUTS

File: dvm_outputs.kicad_sch

Sheets are connected with global labels.

Signals crossing the isolation barrier (3 x HCPL-2631 dual 10 MBd optocouplers):

SCK	MCU -> DVM side	shift clock for both register chains (rising edge shifts)
MOSI	MCU -> DVM side	command bits into 2 x TPIC6C595 (16 open-drain outputs)
LATCH	MCU -> DVM side	low = load 5 x 74HCT165 (40 inputs); rising edge = update command outputs
OEn	MCU -> DVM side	low = enable command outputs (high / LED off = all released)
MISO	DVM side -> MCU	status / BCD data out of the 74HCT165 chain
DRDY	DVM side -> MCU	PRINT command level (SKB pin 34): high = reading complete, data valid

Power: USB VBUS -> AP2112K-3.3 (MCU side). VBUS -> ferrite -> isolated 1 W DC-DC (9 V) -> L78M05 -> +5V_ISO (the 50-way connector has no supply pin, so the isolated logic is powered across the barrier).

Mechanical: 116 x 50 mm, 4 layers. J1 (DD-50 plug) is on the bottom side, everything else on the top side; the board is held by two 4-40 UNC screws through J1's flange holes into the jackposts of SKB.

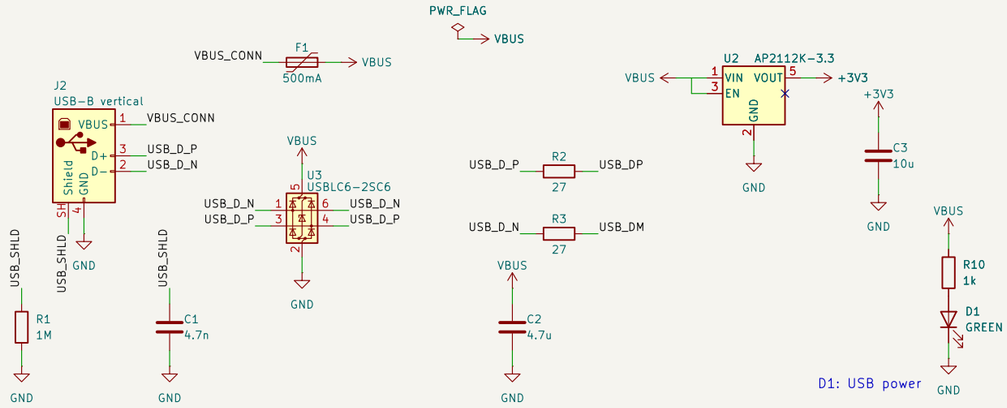
Plugs into SKB (50-way) of the 70754 Parallel BCD Interface Unit
Generated by scripts/gen_sch.py
Solartron 7075 USB interface (RP2354A)

Sheet: /
File: solartron7075_usb_interface.kicad_sch

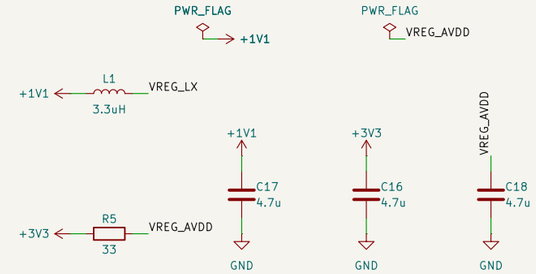
Title: Solartron 7075 opto-isolated USB interface

Size: A4	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/5

USB-B input, ESD protection, 3.3 V LDO (USB / PC ground domain)

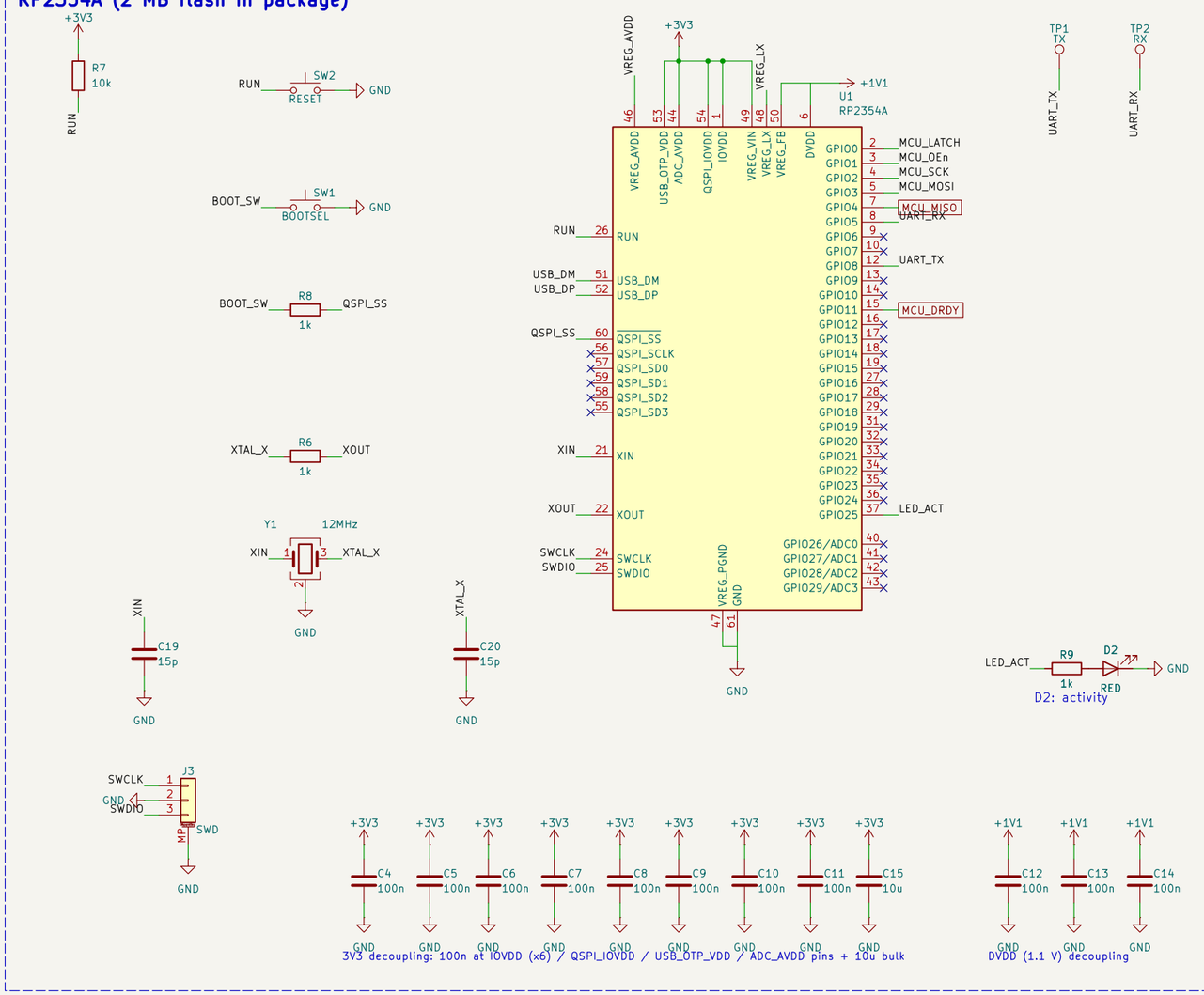


Core 1.1 V switching regulator (RP2350 reference circuit)

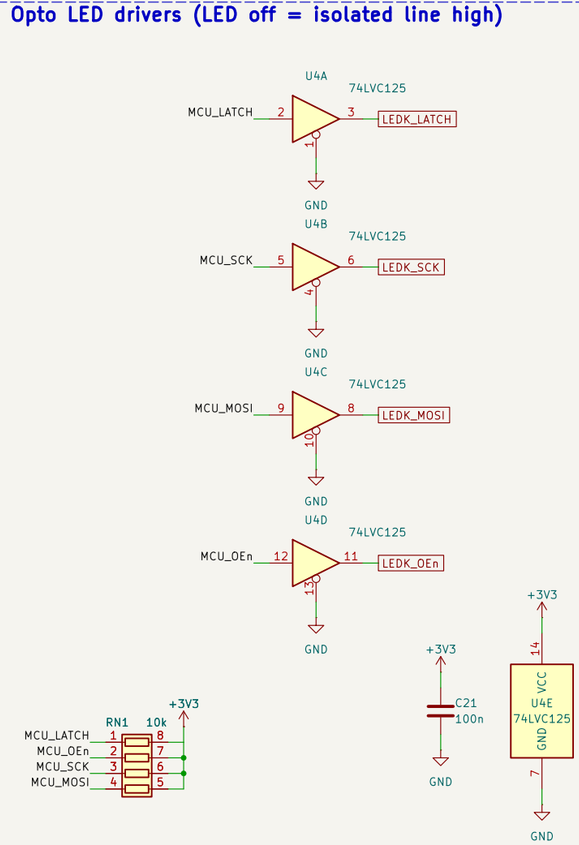


C16 at VREG_VIN, C17 at L1 output, C18 + R5 filter VREG_AVDD.
L1 is polarised: follow the RP2350 datasheet layout.

RP2354A (2 MB flash in package)



Opto LED drivers (LED off = isolated line high)



RN1 pull-ups keep all four opto LEDs off
while the MCU is in reset / unprogrammed.

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Solartron 7075 USB interface (RP2354A)

Sheet: /MCU_USB/
File: mcu_usb.kicad_sch

Title: RP2354A microcontroller, USB and 3.3 V supply

Size: A3	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 2/5

USB / MCU ground domain (GND, +3V3, VBUS)

Isolated multimeter domain (GND_ISO, +5V_ISO)

ISOLATION BARRIER

U5 HCPL-2631
U6 HCPL-2631
U7 HCPL-2631

R11 180
R12 180
R13 180
R14 180
R15 470
R16 470
R17 1k
R18 470
R19 1k
R20 1k
R21 390
R22 390
R23 1k

C22 100n
C23 100n
C24 100n
C25 100n
C26 4.7u
C27 10u
C28 100n
C29 10u
C30 100n

LEDK_SCK
LEDK_LATCH
LEDK_OEn
LEDK_MOSI
LEDK_MISO
LEDK_DRDY
SCK_ISO
LATCH_ISO
OEn_ISO
MOSI_ISO
MISO_ISO
MCU_MISO
MCU_DRDY
PRINT_LEVEL_S

180R: IF = 9 mA typ from 3.3 V

MCU -> multimeter side: SPI-style clock/data, register latch, output enable.
LED off (MCU in reset, USB unplugged) = line high on the isolated side.

Multimeter -> MCU: serial status data and PRINT (data ready) level.
U7 output stage runs from VBUS (5 V); open collectors pull up to 3.3 V.

Uplink LED drivers (TTL-level inputs)

U8A 74AHCT125
U8B 74AHCT125
U8C 74AHCT125
U8D 74AHCT125
U8E 74AHCT125

390R: IF = 8.5 mA typ from +5V_ISO

Isolated supply: USB 5 V -> 9 V isolated DC-DC -> 5 V linear regulator

FB1 120R@100MHz
PS1 B0509S-1WR3
U9 L78M05
PWR_FLAG
D3 GREEN

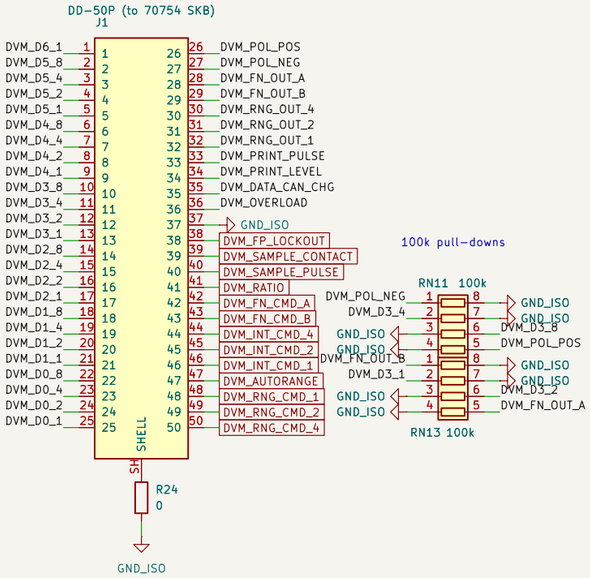
D3: isolated supply OK

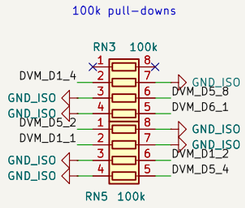
Generated by scripts/gen_sch.py
Solartron 7075 USB interface (RP2354A)
Sheet: /ISOLATION/
File: isolation.kicad_sch

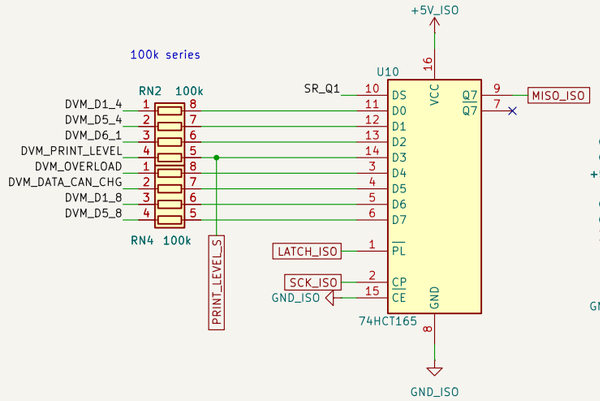
Title: Opto-isolation barrier and isolated 5 V supply

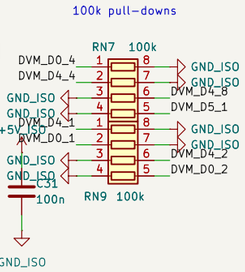
Size: A3	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 3/5

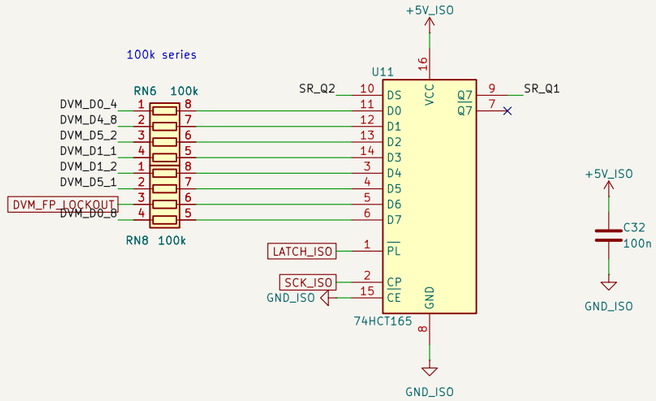
J1 mounts on the BOTTOM side and plugs straight into
the "Systems Interface" socket SKB of the 70754 unit.
Pin numbers = SKB pin numbers (manual page 9.9).

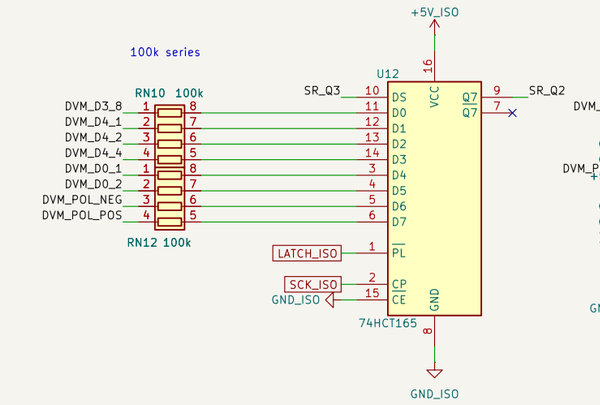


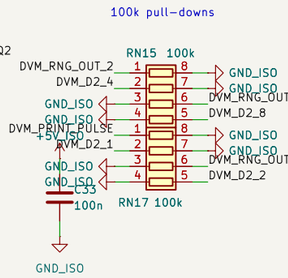


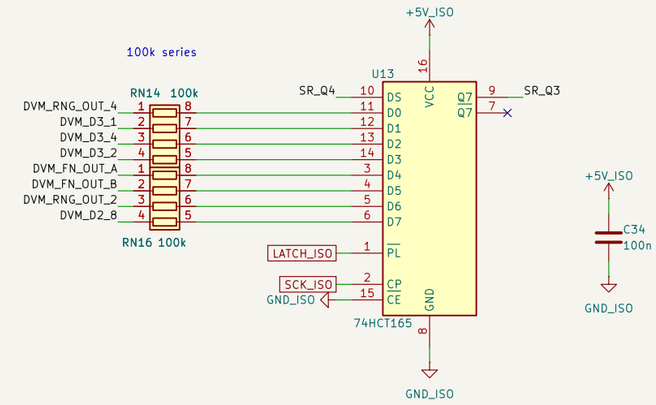




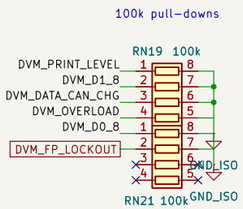


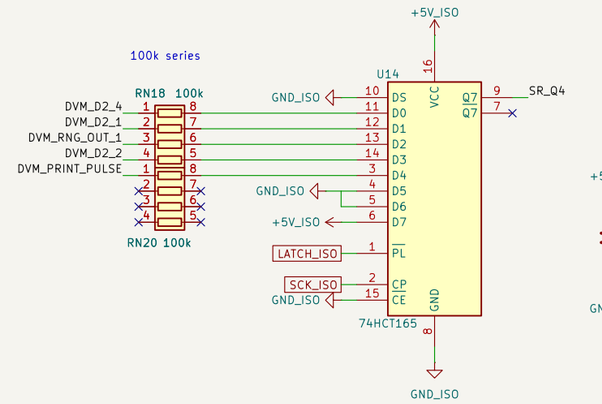






R24: connector shell / jackscrews to GND_ISO.
SKB pin 37 = multimeter EARTH / logic 0.
Remove R24 to float the shell.







Every multimeter output is read through 100k (series) into a 74HCT165 input, with a
100k pull-down on the connector side:
- TTL-compatible thresholds for the 7404 outputs of the 70754 unit
- < 35 uA per line into the input clamp diodes when the isolated supply is off
  (USB unplugged), so the multimeter cannot back-power this board
- defined (all-zero) data when the board is not plugged into the multimeter

Which input reads which SKB pin was chosen for the shortest board layout
(scripts/optimise.py); the firmware uses the table below.

Serial stream at MISO after a LATCH pulse; bit 0 is shifted out first.
Numbers are SKB pins, "=1" / "=0" are hard-wired marker bits (chain length check).

bit 0.. 7	U10 D7..D0 :	2	18	35	36	34	1	3	19
bit 8..15	U11 D7..D0 :	22	38	5	20	21	4	6	23
bit 16..23	U12 D7..D0 :	26	27	24	25	7	8	9	10
bit 24..31	U13 D7..D0 :	14	31	29	28	12	11	13	30
bit 32..39	U14 D7..D0 :	=1	=0	=0	33	16	32	17	15

SKB 38 is the read-back of the FRONT PANEL LOCKOUT line (high = DVM powered).

Generated by scripts/gen_sch.py
Solartron 7075 USB interface (RP2354A)

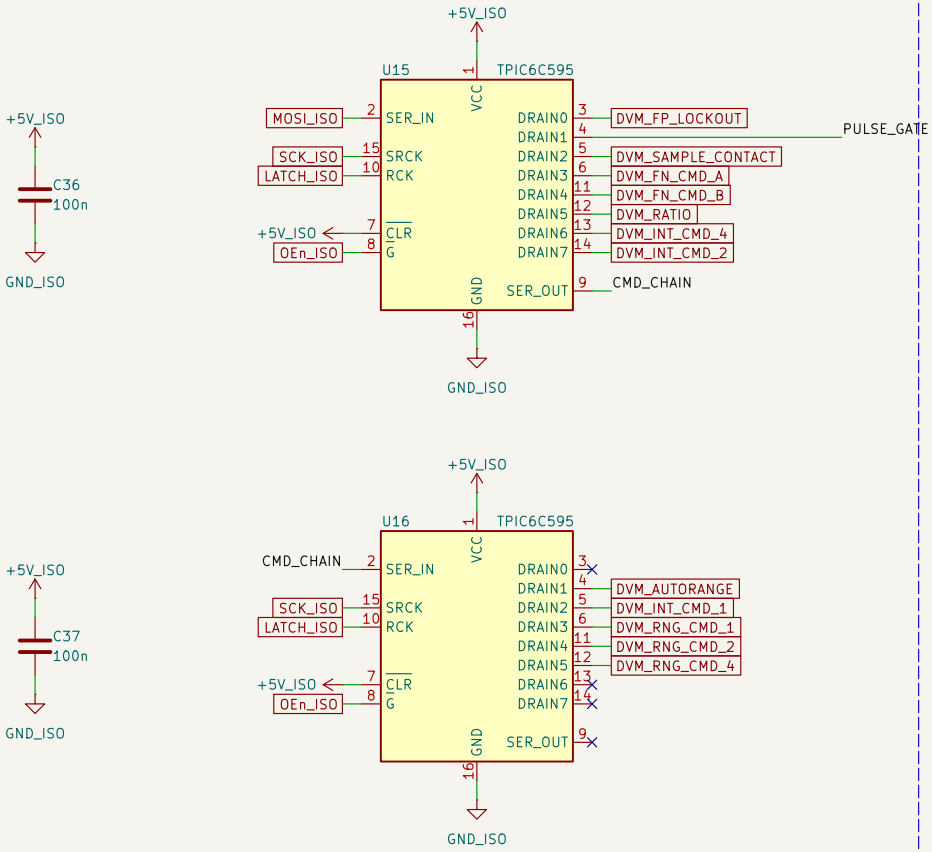
Sheet: /DVM_INPUTS/
File: dvm_inputs.kicad_sch

Title: Multimeter connector and status / BCD data inputs

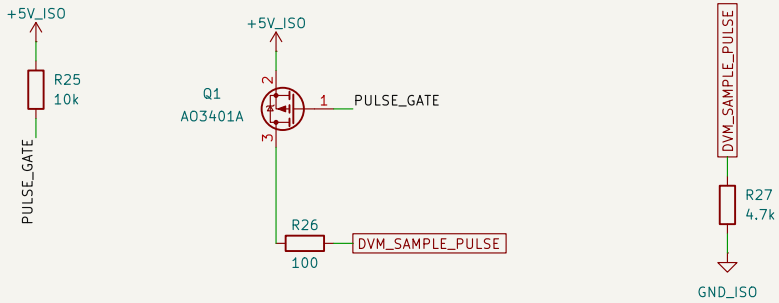
Size: A3	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 4/5

Open-drain command drivers

output off = line released = logic 1 at the multimeter



SAMPLE pulse driver (SKB pin 40: positive pulse > 100 us)



Command word = the last 16 bits clocked into MOSI before LATCH rises (MSB first):

bit 0 -> SKB 38 FRONT PANEL LOCKOUT	bit 8 -> spare
bit 1 -> SKB 40 SAMPLE pulse (via Q1)	bit 9 -> SKB 47 AUTORANGE
bit 2 -> SKB 39 SAMPLE (contact to 0 V)	bit 10 -> SKB 46 INTEGRATION TIME (1)
bit 3 -> SKB 42 FUNCTION	bit 11 -> SKB 48 RANGE (1)
bit 4 -> SKB 43 FUNCTION	bit 12 -> SKB 49 RANGE (2)
bit 5 -> SKB 41 RATIO	bit 13 -> SKB 50 RANGE (4)
bit 6 -> SKB 44 INTEGRATION TIME (4)	bit 14 -> spare
bit 7 -> SKB 45 INTEGRATION TIME (2)	bit 15 -> spare

A 1 turns the open-drain output ON, i.e. pulls the SKB line to logic 0 (SKB 40: a 1 drives the line HIGH through Q1). The multimeter has its own 4.7k pull-ups to +5 V on every command line.

_G (OEn_ISO) is pulled high until the MCU enables it, and all drains are off without isolated power, so an unpowered / unconfigured board leaves the multimeter exactly as if nothing were plugged in.

Generated by scripts/gen_sch.py

Solartron 7075 USB interface (RP2354A)

Sheet: /DVM_OUTPUTS/

File: dvm_outputs.kicad_sch

Title: Multimeter command outputs (open-drain)

Size: A4	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 5/5